



NYC Airbnb Market Explorer

Visualizing spatial and temporal patterns in NYC Airbnb performance using an interactive multi-view dashboard.

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Problem & Motivation

Airbnb market behavior is multidimensional and difficult to understand with static summaries.

Price

+

Demand

+

Location

+

Time

Why static charts are not enough

Tables and one-off charts flatten spatial and temporal variation into a few averages.

Key analytical questions

Where is revenue concentrated? How does performance change over time? How do room types differ?

Visualization goal

Build an interactive system that supports exploration instead of only displaying aggregate statistics.

Dataset

Monthly NYC Airbnb performance data from Kaggle, covering 2021–2026.

35,157

monthly observations

975

unique listings

872

unique hosts

161

ZIP code areas

Key Variables

- Location: latitude, longitude, ZIP code
- Time: monthly records
- Performance: revenue, occupancy, ADR, RevPAR
- Categories: room type, property type

Why visualization is needed

The dataset combines spatial, temporal, categorical, and quantitative variables. Meaningful interpretation requires comparing multiple dimensions together.

System Overview

A coordinated multi-view dashboard for spatial, temporal, and categorical analysis.

Coordinated views

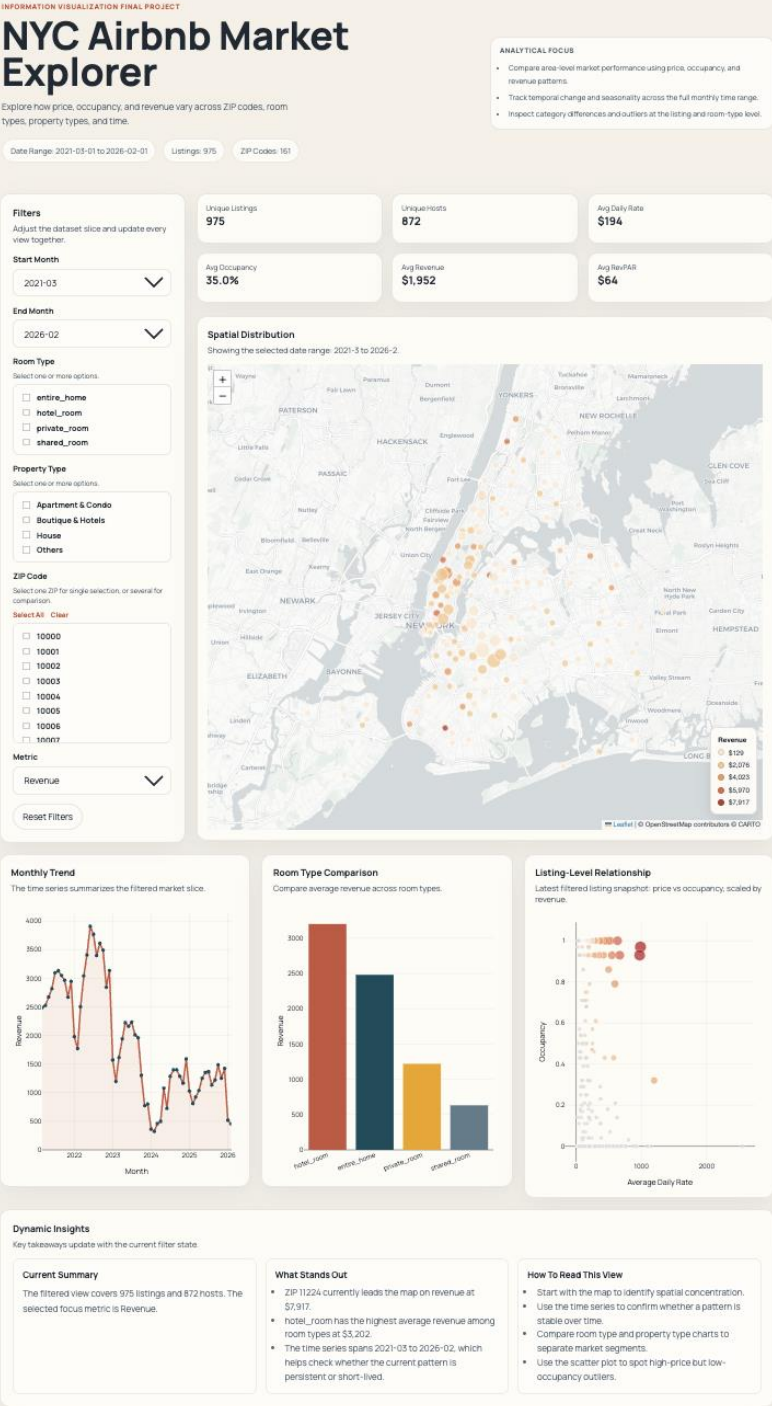
The dashboard integrates map, trend, bar, and listing-level relationship views in one interface.

Interactive filters

Users can adjust time range, room type, property type, ZIP code, and metric to update the dashboard.

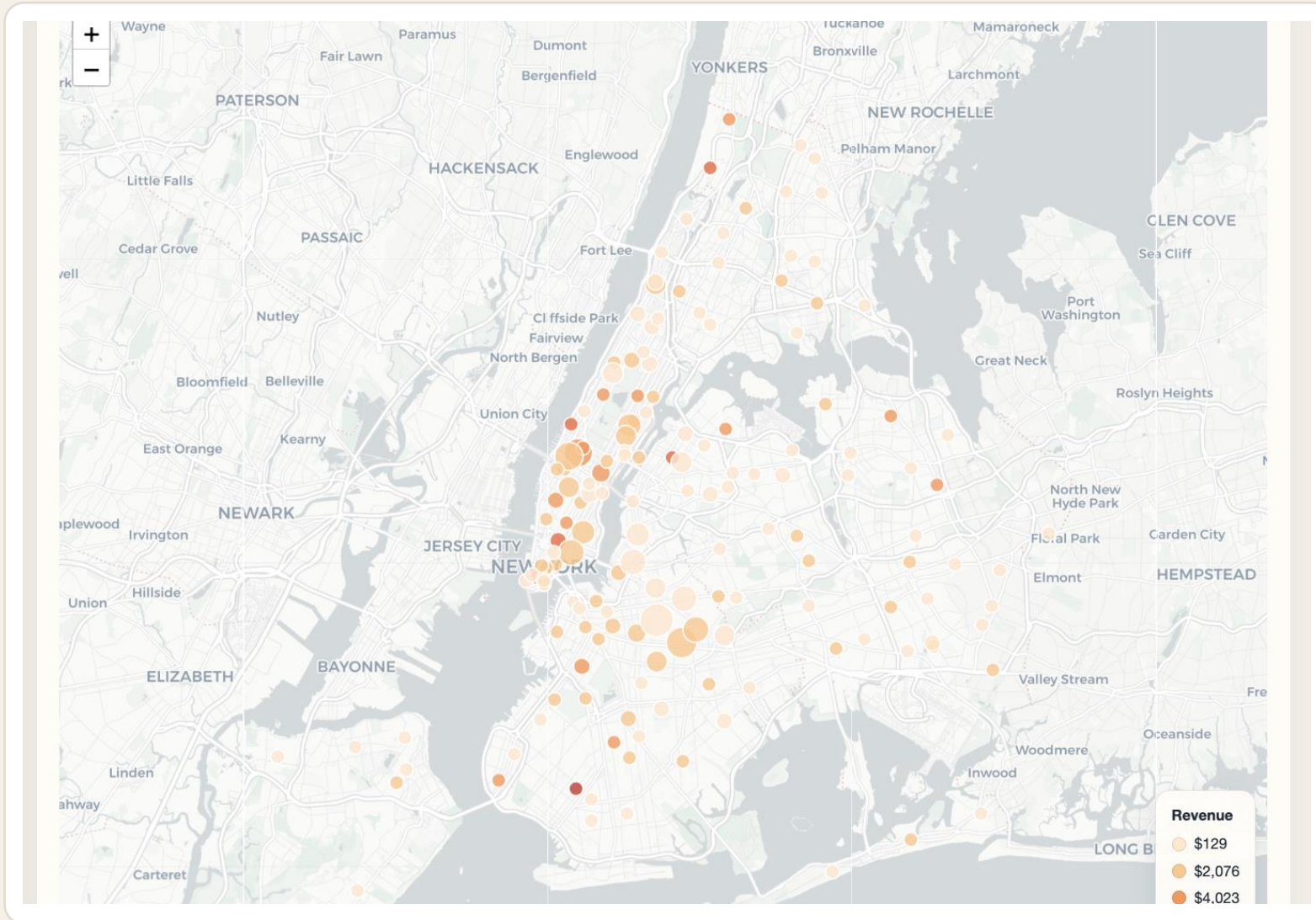
Analysis workflow

Start with the map, compare categories, inspect temporal patterns, then use insights to interpret market behavior.



Spatial Distribution

Revenue is concentrated in selected ZIP-code areas instead of being evenly distributed.



Key Takeaway

- Bubble size reflects revenue
- Color intensity highlights stronger areas
- High-revenue ZIP codes cluster in limited locations

Insight

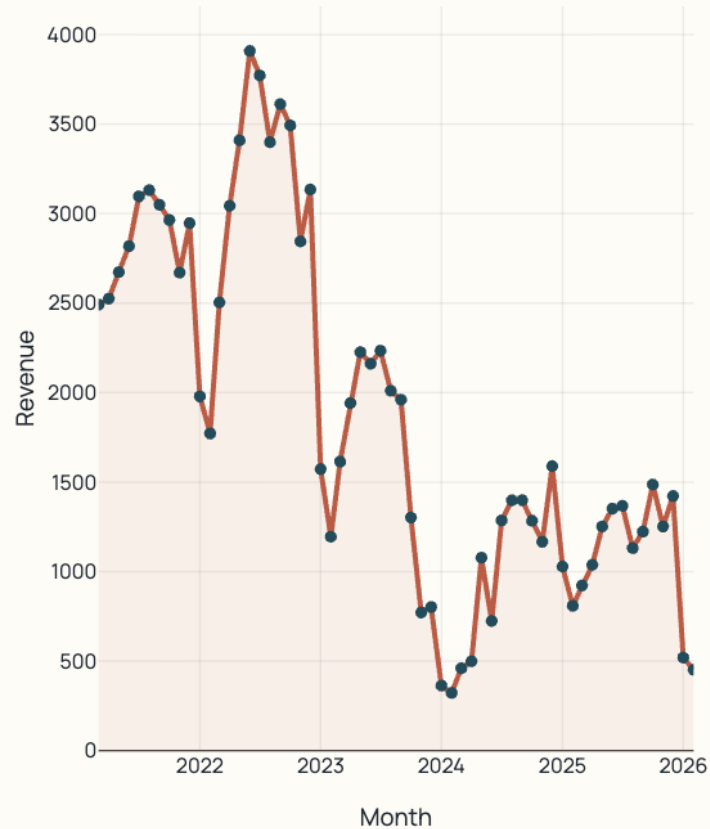
Location plays a major role in earning potential, so Airbnb performance should be evaluated at the local submarket level.

Temporal Trend

Monthly revenue changes substantially over the full data period.

Monthly Trend

The time series summarizes the filtered market slice.



What the chart shows

- Revenue peaks around 2022
- Strong decline through 2023–2024
- Partial recovery afterward
- Performance is not stable over time

Insight

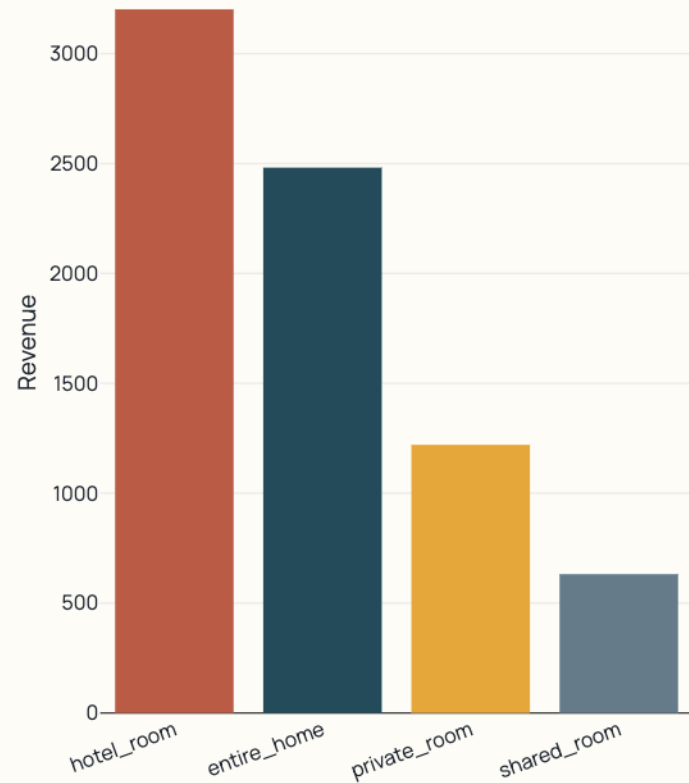
Temporal context matters: a single average would hide major market fluctuations across the 2021–2026 period.

Room Type Comparison

Revenue differs substantially across accommodation formats.

Room Type Comparison

Compare average revenue across room types.



Category Pattern

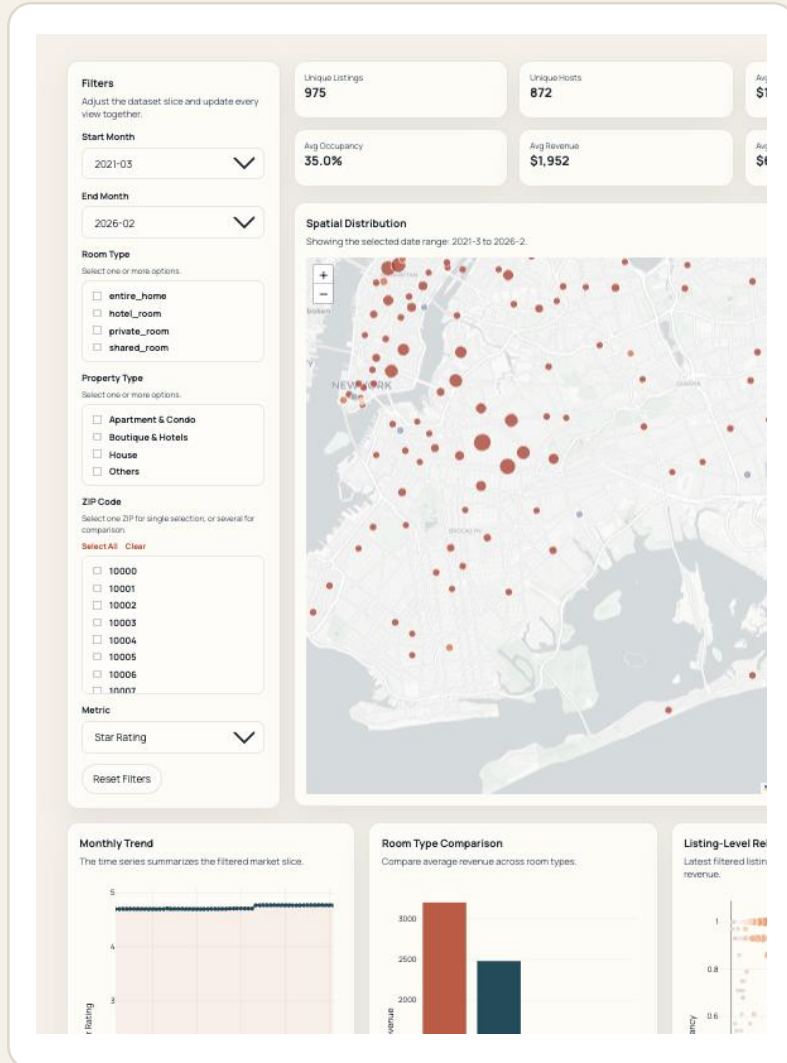
- Hotel rooms have the highest average revenue
- Entire homes also perform strongly
- Private rooms are lower
- Shared rooms have the lowest revenue

Interpretation

Listing category affects market performance, so revenue should not be interpreted only at the city-wide average level.

Interaction & User Experience

Filters support exploratory analysis across space, time, and categories.



Interactive Controls

- Date range filtering
- Room type and property type selection
- ZIP code filtering
- Metric switching
- Dynamic updates across charts

UX refinement

The final dashboard emphasizes a larger spatial view, clearer legend, multi-select filters, and a lighter basemap for readability.

Key Insights

The coordinated views turn raw records into interpretable market patterns.

1. Spatial concentration

Airbnb revenue is concentrated in a limited subset of ZIP-code areas rather than evenly distributed across NYC.

2. Temporal variation

Market performance fluctuates significantly over time, with a strong peak around 2022 and lower levels afterward.

3. Category differences

Room type strongly affects average revenue, with hotel rooms and entire homes outperforming other categories.

Takeaway

Airbnb performance should be interpreted through location, time, and listing category together — not through a single overall average.

Contributions & Live Demo

A working visual analytics system for exploring NYC Airbnb performance.

Core Contributions

- Cleaned and processed multi-year dataset workflow
- Interactive multi-view dashboard
- Spatial, temporal, and category-level insight generation
- Deployed system for browser-based access

Demo Flow

Overview → filter room type → change time range → inspect map and trend → reset

Live Demo URL

<https://nyc-airbnb-market-explorer.vercel.app/>

Thank you!